MILITARY SPECIFICATION

AMPLIFIER, LINEAR POWER: ULTRA HIGH FREQUENCY

1. SCOPE

1.1 This specification establishes the production requirements for one type ultra-high frequency (uhf) solid state linear power amplifier intended for man pack, fixed site or vehicular application, hereinafter referred to as the amplifier.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.
MIL-P-15024	-	Plate, Identification-Information and Marking For Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-16400	-	Electronic Equipment, Naval Ship and Shore, General Specification.
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing Of.

STANDARDS

MILITARY

MIL-STD-108	-	Definition Of and Basic Requirements For Enclosure For Electrical And Electronic Equipment.
MIL-STD-129	-	Marking for Shipment and Storage.

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MIL-STD-242	-	Electronic Equipment Parts (Selected Standards).
MIL-STD-415	-	Test Points And Test Facilities For Electronic Systems And Associated Equipment, Design Standards For.
MIL-STD-449	-	Radio Frequency Spectrum Characteristics, Measurements Of.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipments.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-463	-	Definitions and System of Units, Electromagnetic Interference, Technology.
MIL-STD-470	-	Maintainability Program Requirements (For Systems and Equipment).
MIL-STD-471	-	Maintainability Demonstration.
MIL-STD-756	-	Reliability Prediction.
MIL-STD-781	-	Reliability Test; Exponential Distribution.
MIL-STD-785	-	Reliability Program For Systems And Equipment Development And Production.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-831	-	Test Reports Preparation Of.
MIL-STD-1275	-	Electrical Circuit, 28 Volt DC. Transient Characteristics For Military Vehicles.

HANDBOOKS

MIL-HDBK-217	-	Reliability Stress and Failure Rate Data for Electronic Equipment.

(Copies of specifications, standards and handbooks required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. Prior to submission of any complete production items for acceptance, the contractor shall furnish six first articles for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first articles by the procuring activity shall not relieve the contractor of his obligation to supply amplifiers conforming to this specification. Any changes or deviations of the production items from the first articles shall be subject to the approval of the contracting officer. The first articles shall not be included in the quantity to be delivered to the Government. The first articles shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

3.2 Parts, materials and processes. Parts, material and processes shall be in accordance with MIL-E-16400 except as otherwise specified herein.

3.2.1 Fungus nutrient materials. Fungus nutrient materials shall not be used in the amplifier. Moisture fungus proofing of the overall amplifier shall not be permitted.

3.2.2 Standard parts. Parts shall be selected from MIL-STD-242 wherever possible. All nonstandard parts shall be approved by the procuring activity before delivery of amplifiers.

3.2.3 Solid state devices. Solid state devices shall be utilized exclusively, i.e., vacuum tubes shall not be used.

3.3 Design and construction. This amplifier shall be a uhf broadband solid state linear amplifier capable of linearly amplifying continuous wave or amplitude modulated signals in the 225.00 to 399.95 megahertz (MHz) range. In the transmit mode, the output carrier level shall be 10 watt (W) as specified in 3.6.7 and 3.6.10. In the receive, or standby mode, power on or off, the antenna connection of the amplifier shall be bypassed directly to the radio frequency (rf) input connection. The receive to

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transmit mode change shall be accomplished without the use of an external push-to-talk line. The design shall incorporate protective circuits which will prevent damage to the output amplifier stages in the event of high voltage standing wave ratios (vswr). The amplifier shall be compatible with Radio Sets AN/PRC-41, AN/PRC-41A and AN/PRC-75. The amplifier shall consist of a single unit.

3.3.1 Design approval. The panel layout and operating controls, displays, maintenance facilities and other designs and drawings involving man-machine combinations shall be submitted to the procuring activity for approval prior to amplifier fabrication.

3.3.2 Modes of operation. The amplifier shall have three operating modes which are the transmit mode, the receive or standby mode and the bypass mode. The amplifier shall be in the receive mode when unkeyed with direct current (dc) power on. The amplifier shall be automatically switched to the transmit mode by a signal derived from the drive power. An external key line shall not be required. In the receive mode, the rf input connector shall be connected directly to the rf output connector through the use of cable and mechanical contacts. Solid state switching shall not be used in this switching circuit. The bypass mode shall be the direct connection of the antenna connection to the input connection when the unit is turned off.

3.3.3 Remote operation. The amplifier shall be capable of being operated at a remote site through the use of a coaxial interconnect cable with primary power available at the remote site.

3.3.4 Modular construction. The amplifier shall employ modular design and construction. The modular requirements of MIL-E-16400 shall not apply. The modularity shall be to functional assemblies, i.e., driver, automatic gain control, etc.

3.3.4.1 Modules. The modules shall be capable of being plugged into the main chassis to form a complete working unit. The modules shall be capable of being easily removed from the amplifier either singly or in any combination without unsoldering any connections. The ability to improperly replace modules shall be eliminated by keying connectors and modules. No loose parts which can be lost during disassembly shall be used within the modules. All such parts shall be permanently attached (i.e., captivated screws) as an integral part of the amplifier.

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3.3.4.2 Module identification. Each module shall be so constructed and marked that replacement by similar appearing modules shall not be possible.

3.3.5 Test points. Test points shall be incorporated for major circuit maintenance/repair test and shall be in accordance with MIL-STD-415.

3.3.6 Controls. The following controls shall be provided:

3.3.6.1 Frequency selection controls. The amplifier shall not use frequency selection controls; it shall use broadband techniques and cover the entire 225- to 400-MHz band.

3.3.6.2 Power control. A power ON-OFF switch shall be provided for control of power to the amplifier circuitry. When the switch is in the OFF position, the rf input shall be connected directly to the rf output.

3.3.6.3 Primary power connection. The primary power termination on the amplifier shall be a watertight multipin receptacle with such external connections as to allow connection to either vehicle or battery power. A chain captivated cap for this receptacle shall be provided.

3.3.6.4 Antenna connections. The input and output coaxial connectors shall be waterproof type N with chain captivated caps.

3.3.6.5 Power output indicator. A visual indicator shall be provided which will display "HIGH" or "LOW" information to the radio operator. The indicator shall be a MINELCO type B1521-( ), or equal.

3.3.7 Input power. The amplifier shall be designed to perform as specified herein when operated from any power source with the following characteristics without necessitating internal changes in the equipment:

(a) Rechargeable battery

1. BB451/U

2. BB626/U

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(b) Nominal 27.5-volt (V) dc vehicle system within the limits of 22 to 32 V meeting the requirements of MIL-STD-1275.

3.3.8 Fuse protection. The amplifier shall provide an externally accessible protective fuse in the dc power input line. A spare fuse shall also be provided externally.

3.3.9 Reverse polarity protection. The amplifier shall contain protective circuitry to prevent damage which may result from an improperly polarized dc voltage of 32.0 V.

3.3.10 Watertightness. The amplifier shall be enclosed in a submersible case in accordance with MIL-STD-108. The basic components shall be submersible when assembled in operating conditions.

3.3.11 Mechanical stability. All controls and adjustments shall be designed so their position of adjustments will not be altered when the amplifier is subjected to the service conditions specified in 3.6.3.

3.3.12 Cooling. The amplifier shall be cooled using using natural means. Blowers, or other mechanical cooling devices shall not be used.

3.4 Reliability.

3.4.1 Mean time between failure (MTBF). The amplifier shall possess MTBF of no less than 1,500 hours when tested in accordance with 4.6.5. This is the specified MTBF as defined in MIL-STD-781. A failure shall be defined as any degradation below this minimum MTBF.

3.4.2 Reliability program. The required reliability shall be obtained through a reliability program performed in accordance with MIL-STD-785 1/ with exceptions or additions as follows:

(a) The contractor shall perform an analysis of the proposed design of the amplifier to determine compatibility with the required MTBF. A failure rate shall be assigned to each part in accordance with data presented in MIL-HDBK-217.

1/ Quantitative reliability shall be predicted in accordance with MIL-STD-756, type II.

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Parts not included in MIL-HDBK-217 shall be assumed to possess the failure rate of the most similar part in MIL-HDBK-217. Where this is unrealistic, any valid existing data shall be used upon approval of the procuring activity. The analysis shall include a comparison of the predicted with the required MTBF. Where the predicted figure is less than the requirement, the contractor shall accomplish such changes in design as are necessary to raise the predicted MTBF to the required value.

(b) The reliability analysis shall be revised periodically to consider design changes or improved data on part stress levels.

(c) As the fabrication is completed so that normal amplifier performance is possible, the contractor shall prepare and submit to the procuring activity a final prediction of the reliability of the amplifier. If the prediction indicates a lesser reliability than that required, the contractor shall accomplish such changes in design, part application and part stress level as are necessary to yield a MTBF of the required level.

3.5 Maintainability. The amplifier shall meet the following maintainability requirements.

3.5.1 Maintainability program. The contractor shall prepare, for approval, a maintainability program plan in accordance with the requirements of MIL-STD-470.

3.5.2 Mean time to repair (MTTR). The MTTR of the amplifier shall not exceed 30 minutes to the module level.

3.6 Performance. The performance of the amplifier shall be as specified herein when operated with the AN/PRC-41, AN/PRC-41A and AN/PRC-75 radio sets.

3.6.1 Equipment life. The amplifier shall exhibit a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts.

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3.6.2 Standard conditions. The following shall be used as a basis for establishing normal performance requirements and for conducting tests of the amplifiers:

(a)	Barometric Pressure	28 to 32 inches mercury
(b)	Temperature	15°C to 35°C
(c)	Humidity	50 percent to ±30 percent
(d)	Supply Voltage	27.5 ±0.5 direct-current volts (Vdc)

3.6.3 Service conditions. The amplifier shall be capable of operating under the following service conditions:

(a)	Ambient temperature range	40°C to +65°C
(b)	Ambient humidity range	0 to 100 percent relative
(c)	Vibration	Method 514, Procedure 1 Curve W of Figure 514-5, MIL-STD-810
(d)	Shock	Method 516, Procedure 1, and Procedure 5, MIL-STD-810
(e)	Leakage	Method 512, Procedure 1, MIL-STD-810
(f)	High Temperature	Method 501, Procedure 1, MIL-STD-810
(g)	Low Temperature	Method 502, Procedure 1, MIL-STD-810
(h)	Temperature Shock	Method 503, Procedure 1, MIL-STD-810
(i)	Humidity	Method 507, Procedure 1, MIL-STD-810
(j)	Altitude	Method 504, Procedure 1, MIL-STD-810

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(k)	Salt Fog	Method 509, Procedure 1, MIL-STD-810
(l)	Fungus	Method 508, Procedure 1, MIL-STD-810

3.6.4 Storage conditions. The amplifier shall be capable of being stored under the following conditions without degradations:

(a)	Ambient temperature range	-62°C to +65°C
(b)	Altitude	Sea level to 70,000 feet

3.6.5 Frequency coverage. The amplifier shall be capable of operating at any frequency within the 225.00- to 399.95-MHz range.

3.6.6 Radio frequency input impedance.

3.6.6.1 Transmit mode. The rf input impedance in the transmit mode shall be nominally 50 ohms with a vswr of not more than 1.5 to 1.

3.6.6.2 Receive mode. In the receive mode, with rf output connector terminated with a 50-ohm resistive load, the input impedance shall be nominally 50 ohms with a vswr of not more than 1.5:1.

3.6.7 Radio frequency input. The amplifier shall provide rated power output with rf input signals falling in the range of 0.500 W to 5.0 W, average, modulated 90 percent 300-hertz to 20-kilohertz sine wave.

3.6.8 Radio frequency output load impedance. The amplifier shall be capable of working into an rf load impedance of 50 ohms nominal, with a vswr of not more than 4.0 to 1.

3.6.9 Radio frequency power output. The amplifier shall deliver the defined carrier power to a 50-ohm resistive load throughout the frequency range specified herein.

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3.6.10 Radio frequency power output variation. Under standard conditions, the carrier power output of the amplifier shall not vary more than 1 decibel (dB) from 10 W across the 225-MHz to 400-MHz range. Under service conditions, the power output shall not vary more than ±0.5 dB from standard conditions measurement.

3.6.11 Harmonic and spurious outputs. Harmonically related spurious outputs shall be at least 60 dB below, and all other spurious outputs shall be at least 80 dB below, the nominal carrier output level when measured in accordance with MIL-STD-449 and when the amplifier is driven by a spurious free signal generator.

3.6.12 Envelope distortion. The linearity of the amplifier shall be such that with an input signal 90 percent modulated and containing not more than 10 percent total harmonic distortion, the total harmonic distortion of the detected rf output envelope shall not exceed 20 percent.

3.6.13 Amplitude carrier shift. The linearity of the amplifier shall be such that with an input signal 90 percent modulated and containing not more than 10 percent amplitude carrier shift, the total amplitude carrier shift at the rf output envelope shall not exceed 20 percent.

3.6.14 Radio frequency intermodulation.

3.6.14.1 Carrier on condition. The amplitude of each rf intermodulation product shall be at least 30 dB below the amplitude of an interfering signal fed into the equipment output connector at 20 dB below the amplifier output level and spaced ±1 MHz from the output frequency of the transmitter when measured in accordance with the test methods of MIL-STD-449.

3.6.14.2 Receive mode. With the amplifier in the receive mode, the receiver intermodulation or cross modulation products of the AN/PRC-41, AN/PRC-41A and AN/PRC-75 radio sets shall not be degraded.

3.6.15 Transmit-receive switching. The amplifier shall be automatically switched from the receive mode to transmit mode when the rf signal source is within the range specified in 3.6.7. A reduction of drive power or any interruption of primary power shall return the amplifier to the receive mode.

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3.6.16 Voltage standing wave ratio protection. The equipment shall not be damaged when working into loads of 4:1 vswr. Protective circuits which will automatically switch the amplifier to the receiver mode in the event of an open or shorted antenna shall be incorporated in the design. The amplifier shall also be automatically switched to the receive mode if the forward output power drops below 5 W average.

3.6.17 Power consumption. The power consumption with standard input voltage (27.5 Vdc), shall not be greater than the amount listed below for the indicated conditions using a transmitter dummy load of 50 ohms.

(a) Transmit (90 percent sine wave modulation) - 100 W

(b) Receive mode - 3 W

3.6.18 Duty cycle. The amplifier shall be capable of continuous transmit at 90 percent modulation and rated output power for 2 hours under standard environmental conditions. Above +50°C, the duty cycle shall be 1 minute transmit and 1 minute receive. The amplifier shall meet all performance requirements specified herein when operating under standard conditions and shall be capable of constant and reliable performance under any probable combination of service conditions.

3.7 Radio interference suppression. The amplifier shall meet the interference control limits of MIL-STD-461 using the test methods of MIL-STD-462. Terms and definitions shall be in accordance with MIL-STD-463.

3.8 Dimensions. The amplifier dimensions shall not exceed 5-1/8 by 6 by 9 inches excluding connectors and mounting brackets.

3.9 Weight. The amplifier weight shall not exceed 9 pounds.

3.10 Finish and color. The amplifier shall be painted and treated in accordance with MIL-T-704, type A; color shall match Marine Corps green, No. 23 (see 6.3).

3.11 Identification and marking. The amplifier shall contain the identification name plates and instruction plates in accordance with MIL-E-16400.

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3.11.1 Nameplates. Nameplates and nomenclature shall be in accordance with MIL-P-15024.

3.11.2 Panel markings. Each connector and control shall be marked to aid operating personnel.

3.12 Workmanship. The amplifier covered by this specification, including all parts and accessories, shall be manufactured in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Production (see 4.4).

(c) Sample (see 4.5).

4.3 First article inspection. The six first articles shall be subjected to tests indicated for first article in Table I. The Government representative present will select which amplifier is to be subjected to each test.

4.3.1 First article test reports. When the contractor conducts the preliminary tests, he shall, after completion of the tests, prepare a test report.

4.4 Production inspection. All amplifiers shall be subjected to the production tests indicated in Table I.

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Table I - Tests

Test	Reqt Para.	Test Para.	First Article Tests	Prod. Tests	Sample Tests	Test No.
Mechanical		4.6.1	X	X	X	1
Electrical		4.6.2				
Continuity		4.6.2.1	X	X	X	2
Electrical Measurement		4.6.2.2	X	X	X	3
Standard Performance Tests		4.6.3				
Input Power	3.3.7, 3.6.17		X	X	X	4
Radio Frequency Power Output	3.6.9, 3.6.10		X	X	X	5
Radio Frequency Input and Mod Capability	3.6.7		X	X	X	6
Power Output Indicator	3.3.6.5		X	X	X	7
Frequency Range	3.6.5		X	X	X	8
Spurious and Harmonic Outputs	3.6.11		X	X	X	9
Voltage Standing Wave Ratio Protection	3.6.16		X	X	X	10
Radio Frequency Input Impedance	3.6.6		X	X	X	11
Radio Frequency Output Load	3.6.8		X	X	X	12

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Table I - Tests (continued)

Test	Reqt Para.	Test Para.	First Article Tests	Prod. Tests	Sample Tests	Test No.
Envelope distortion	3.6.12		X	X	X	13
Amplitude Carrier Shift	3.6.13		X	X	X	14
Transmit-receive Switching	3.6.15		X	X	X	15
Performance Tests		4.6.4				
Radio Frequency Intermodulation	3.6.14		X		X	16
Modes of Operation	3.3.2		X		X	17
High and Low Voltage Operation	3.3.7 3.3.9		X		X	18
Battery Operation	3.3.7		X		X	19
Duty Cycle	3.6.18		X		X	20
Radio Interference Suppression	3.7		X		X	21
Reverse Polarity Protection	3.3.9		X		X	22
Fuse Protection	3.3.8		X		X	23
Reliability	3.4	4.6.5	X		X	24
Environmental Tests	3.6.3	4.6.6				
Low Temperature		4.6.6.5	X		X	25
High Temperature		4.6.6.4	X		X	26

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Table I - Tests (continued)

Test	Reqt Para.	Test Para.	First Article Tests	Prod. Tests	Sample Tests	Test No.
Altitude		4.6.6.8	X		X	27
Temp. Shock		4.6.6.6	X		X	28
Leakage		4.6.6.3	X		X	29
Dust		4.6.6.11	X		X	30
Shock		4.6.6.2	X		X	31
Bench Handling		4.6.6.2	X		X	32
Vibration		4.6.6.1	X		X	33
Fungusproofing		4.6.6.10	X		X	34
Humidity		4.6.6.7	X		X	35
Salt Fog		4.6.6.9	X		X	36
Electromagnetic Interference Tests		4.6.7	X		X	37
Maintainability Test	3.5	4.6.8	X		X	38

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4.5 Sample inspection. During the production phase, sample tests shall be conducted. The sample tests shall consist of all tests indicated in Table I. A minimum of one sample test amplifier shall be randomly selected by the cognizant Government inspector from each block of cumulative quantity produced as indicated below:

Sample Number	Cumulative Quantity Produced	Quantity to be Tested
1	1-10	1
2	11-100	1
3	101-200	1
		One additional sample for each 100 amplifiers

4.6 Tests. Test plans describing in detail first article, production and sample testing shall be prepared by the contractor in accordance with the requirements of MIL-STD-831 and submitted to the contracting officer for approval prior to initiation of testing within the scope of the test plan. Approval of the test plan by the contracting officer or his representative is required prior to commencement of the testing under the plan. The test plans shall include all test procedures deemed necessary and practicable by the contractor to demonstrate compliance with this specification.

4.6.1 Mechanical inspection. The amplifier shall be given a thorough mechanical and visual inspection to determine that the quality of all materials and workmanship is in compliance with the requirements of this specification and MIL-E-16400. Particular attention shall be given to the following:

(a) Completeness

(b) Nameplates, identification markings and labels

(c) Ease of operation of connector mating, thumb screws, controls and switches and any other part intended for adjustment, movement or operation.

(d) Finishes

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(e) Welded joints

(f) Check of solder joints

(g) The fit of components in their respective positions

(h) Check of mounting means

(i) Check of safety features

(j) Fastening and securing devices or parts

(k) Accessibility of components and parts for servicing

(l) Cable run between components including plugs and receptacles

(m) Fit of sealing gasket on cover and wear on gasket

(n) Check of all connector fittings

(o) Overall dimensions

(p) Weight

(q) Other visual defects

4.6.2 Electrical tests. The amplifier shall be given all electrical tests necessary to confirm that all circuits are inherently sound and in compliance with the requirements of this specification. The electrical tests shall include, but not necessarily be restricted to, the following.

4.6.2.1 Continuity. Each electrical component and each cord and cable shall be given a continuity test as necessary to ascertain if it is wired and connected correctly, good electrical contact is obtained, and if it is wired in accordance with the wiring diagrams to be delivered under the contract.

4.6.2.2 Electrical measurements. Measurements of important operating voltages at module pins, connector terminals, and all electrical junctions throughout the equipment shall be checked and recorded for conformance to those shown on the circuit labels, schematic drawings,

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wiring diagrams and troubleshooting charts. This shall be done with all controls set for normal operation of the amplifier. The circuit labels, schematic drawings, wiring diagrams and troubleshooting charts shall be provided as part of the test procedures. Circuit labels shall include voltage levels and waveshapes. Resistance and reactance measurements from the above points to ground shall be made with all controls set for normal operation of the amplifier, but with no power applied. Instruments of the vacuum-tube-voltmeter-type or solid state equivalent shall be used for these tests. Conclusive agreement shall be reached between the contractor and the procuring activity on the specific values of voltage, resistance, et cetera, in the above mentioned tests.

4.6.3 Standard performance tests. All amplifiers shall be given the standard performance tests indicated in Table I to confirm that the amplifiers meet the performance specified herein. The standard performance tests shall be performed in conjunction with each environmental test.

4.6.4 Performance tests. First article and sample production amplifiers shall be given the performance tests indicated in Table I to confirm that the amplifiers meet the performance specified herein.

4.6.5 Reliability tests. The reliability specified in 3.4 shall be demonstrated as specified herein.

4.6.5.1 First article test. This test shall be performed in accordance with MIL-STD-781.

4.6.5.1.1 Test level. Test level F of MIL-STD-781, modified with temperature limits of -40°C to +65°C, shall apply.

4.6.5.1.2 Sample size. Four first article units shall be subjected to the reliability test.

4.6.5.1.3 Test plan. Test Plan III of MIL-STD-781 shall apply.

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4.6.5.1.4 Duty cycle. For purposes of reliability testing, the duty cycle shall be 1 minute carrier on-1 minute carrier off. The amplifier shall be turned off during the temperature cooling cycle. The amplifiers shall be driven with 3 W +10 percent -0 percent in the carrier on condition.

4.6.5.1.5 Voltage cycling. The voltage cycling shall be in accordance with MIL-STD-781.

4.6.5.2 Production test. Each sample production unit selected for testing shall be tested in accordance with MIL-STD-781.

4.6.5.2.1 Test level. Test level shall be same as 4.6.5.1.1.

4.6.5.2.2 Test plan. Test plan shall be same as 4.6.5.1.3.

4.6.5.3 Analysis of failures. Failures shall be analyzed in accordance with MIL-STD-781.

4.6.6 Environmental tests. The amplifier shall be subjected to the following environmental tests performed in accordance with MIL-STD-810 in the sequence specified by Table I, group 1 E, of MIL-STD-810. To determine compliance with the requirements of 3.6, the standard performance tests of Table I shall be performed.

4.6.6.1 Vibration. The amplifier shall be subjected to a vibration test in accordance with Method 514, procedure 1, curve W, of Figure 514-5 of MIL-STD-810.

4.6.6.2 Shock test. Shock test shall be in accordance with Method 516, Procedures I and V of MIL-STD-810. The shock pulse shape shall be in accordance with Figure 516-2, amplitude a and time duration c. The amplifier shall be turned on and in the receive mode. Any failures shall be repaired in accordance with 3.5.2.

4.6.6.3 Leakage. Leakage test shall be in accordance with Method 512, procedure 1 of MIL-STD-810.

4.6.6.4 High temperature. High temperature test shall be in accordance with Method 501, procedure 1, of MIL-STD-810. Test operation of amplifier at +65°C using 22, 27.5 and 32 Vdc source. Maximum temperature to be +65°C.

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4.6.6.5 Low temperature. Low temperature test shall be in accordance with Method 502, procedure 1, of MIL-STD-810 except that after 72 hours at -62°C, the temperature will be raised to -40°C to test operation using 22, 27.5 and 32 Vdc source.

4.6.6.6 Temperature shock. Temperature shock test shall be in accordance with Method 503, Procedure 1, of MIL-STD-810 with the following exceptions:

(a) High temperature shall be +65°C.

(b) The test shall be concluded at the end of three cycles.

4.6.6.7 Humidity. Humidity test shall be in accordance with Method 507, Procedure 1, of MIL-STD-810 with the following exceptions:

(a) Maximum temperature shall be +65°C.

(b) The test will be concluded at the end of three cycles.

(c) Moisture may be removed by wiping.

4.6.6.8 Altitude. Altitude test shall be in accordance with Method 504, Procedure 1, of MIL-STD-810 equipment Class 1B, except the maximum temperature shall be +65°C.

4.6.6.9 Salt fog. Salt fog test shall be performed with amplifier in test chamber in operating condition in accordance with Method 509, Procedure 1, of MIL-STD-810.

4.6.6.10 Fungusproofness. Fungusproofness shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810.

4.6.6.11 Dust. Dust test shall be in accordance with Method 510, Procedure 1, of MIL-STD-810.

4.6.7 Electromagnetic interference test. The amplifier shall be subjected to test defined in MIL-STD-461 equipment type 1B.

4.6.8 Maintainability test. The maintainability as specified in 3.5 shall be demonstrated as specified herein.

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4.6.8.1 Module replacement. This test shall be performed in accordance with Method II of MIL-STD-471.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The uhf linear power amplifier is intended for use as a range extender in the 225-to 400-MHz band. It is intended as an add-on amplifier to the AN/PRC-41, AN/PRC-41A, AN/PRC-75 and other uhf lower power radios. As a range extender it will increase the air-ground communication range. The configuration is such as to permit use in man-pack, vehicular or fixed site installations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) Levels of preservation, packaging and packing (see 5.1).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions C-03, E-14, P-01, R-02, R-05, R-07, T-01, T-02, T-03, T-05, T-06, T-07, T-08, T-09, T-10 and V-11 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

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6.4 First article instructions. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. Selected first article tests will be witnessed by a representative of the procuring activity. Tests not witnessed by a representative of the procuring activity shall be witnessed and certified by the appropriate Defense Contract Administration Services District representative. Approval of the first article shall be by the contracting officer.

Preparing activity:
Navy - MC
Project No. 5820-N490

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